

Solutions to Homework #9

Problem 1

- (a) Let $V(z) = (1 + z^{-1})^3(1 + z)^3$. Then $v[n]$ can be computed in Matlab as follows.

```
v = conv([1 2 1],[1 2 1]);
v = conv([1 2 1], v);
```

Note that $v[k]$ in Matlab corresponds to $v[k-4]$ for $k = 1, 2, \dots, 7$. Let $q[n] = q_2\delta[n+2] + q_1\delta[n-1] + q_0\delta[n] + q_1\delta[n-1] + q_2\delta[n-2]$. Then $A_0(z) = Q(z)V(z)$. Since $a_0[2m] = \delta[m]$, we have

$$\begin{aligned}\sum_{k=-2}^2 v[4-k]q[k] &= a_0[4] = 0, \\ \sum_{k=-2}^2 v[2-k]q[k] &= a_0[2] = 0, \\ \sum_{k=-2}^2 v[-k]q[k] &= a_0[0] = 1.\end{aligned}$$

Furthermore, we have

$$\begin{bmatrix} q_2 \\ q_1 \\ q_0 \\ q_1 \\ q_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} q_0 \\ q_1 \\ q_2 \end{bmatrix}.$$

Therefore $[q_0, q_1, q_2]^\top$ is computed in Matlab as follows.

```
T = [zeros(1,3) v(7:-1:6) ; 0 v(7:-1:4) ; v(6:-1:2)];
E = [0 0 1; 0 1 0; 1 0 0; 0 1 0; 0 0 1];
q = (T*E)\[0; 0; 1];
```

- (b) The spectral factorization finds the roots of $Q(z)$ inside the unit circle. This can be done as follows.

```
arr_zeros = roots([q(3:-1:2) ; q])
idx = find(abs(arr_zeros) <= 1)
arr_zeros = arr_zeros(idx)
r0 = sqrt(sum([q(3:-1:2) ; q])/(prod(abs(1-arr_zeros))^2))
```

Here we computed r_0 by

$$|R(e^{j0})|^2 = r_0^2 \cdot |1 - \alpha_1|^2 \cdot |1 - \alpha_2|^2 = Q(e^{j0}) = q_2 + q_1 + q_0 + q_1 + q_2,$$

where α_1 and α_2 denote the roots of $R(z)$.

- (c) Next, given $R(z)$, we compute $G_0(z)$ by $G_0(z) = (1 + z^{-1})^3 R(z)$, which is implemented in Matlab as follows.

```
r0 = sqrt(sum([q(3:-1:2) ; q])/(prod(abs(1-arr_zeros))^2))
g0 = conv([1 3 3 1],r0*[1 -sum(arr_zeros) prod(arr_zeros)])
```

- (d) One can derive the relation among filters by computing the inverse z -transform of the given identities. (This is left as exercises.) Then we obtain the followings.

```
g0 =
    0.3327    0.8069    0.4599   -0.1350   -0.0854    0.0352
g1 =
    0.0352    0.0854   -0.1350   -0.4599    0.8069   -0.3327
h0 =
    0.0352   -0.0854   -0.1350    0.4599    0.8069    0.3327
h1 =
   -0.3327    0.8069   -0.4599   -0.1350    0.0854    0.0352
```

- (e) By the construction, all filters have unit norm.
- (f) Recall that the system has perfect reconstruction, that is, $y[n]$ is just a delayed version of the input $x[n]$. Therefore, it suffices to show the identity where $x[n]$ is substituted by $y[n]$. In class, we have derived that

$$\begin{aligned} y[n] &= \left(\sum_{k=-\infty}^{\infty} g_0[n-2k]a[k] \right) + \left(\sum_{k=-\infty}^{\infty} g_1[n-2k]d[k] \right) \\ &= \sum_{k=-\infty}^{\infty} (g_0[n-2k]a[k] + g_1[n-2k]d[k]). \end{aligned}$$

Let $\langle \cdot, \cdot \rangle$ denotes the dot product of two discrete-time signals. Then the energy of $y[n]$ is

given by

$$\begin{aligned}
& \langle y[n], y[n] \rangle \\
&= \left\langle \sum_{k=-\infty}^{\infty} (g_0[n-2k]a[k] + g_1[n-2k]d[k]), \sum_{\ell=-\infty}^{\infty} (g_0[n-2\ell]a[\ell] + g_1[n-2\ell]d[\ell]) \right\rangle \\
&= \sum_{k=-\infty}^{\infty} \sum_{\ell=-\infty}^{\infty} \left\langle (g_0[n-2k]a[k] + g_1[n-2k]d[k]), (g_0[n-2\ell]a[\ell] + g_1[n-2\ell]d[\ell]) \right\rangle \\
&= \sum_{k=-\infty}^{\infty} \sum_{\ell=-\infty}^{\infty} \left(a[k]a[\ell] \langle g_0[n-2k], g_0[n-2\ell] \rangle + a[k]d[\ell] \langle g_0[n-2k], g_1[n-2\ell] \rangle \right. \\
&\quad \left. + d[k]a[\ell] \langle g_1[n-2k], g_0[n-2\ell] \rangle + d[k]d[\ell] \langle g_1[n-2k], g_1[n-2\ell] \rangle \right).
\end{aligned}$$

Note that we used Daubechie's orthogonal design. Therefore we have

$$\begin{aligned}
g_0[n-2k], g_0[n-2\ell] &= \delta[k-\ell], \\
g_1[n-2k], g_1[n-2\ell] &= \delta[k-\ell], \\
g_0[n-2k], g_1[n-2\ell] &= 0, \\
g_1[n-2k], g_0[n-2\ell] &= 0
\end{aligned}$$

for all k and ℓ . Therefore, the expression of the energy of $y[n]$ simplifies to

$$\begin{aligned}
\langle y[n], y[n] \rangle &= \sum_{k=-\infty}^{\infty} \sum_{\ell=-\infty}^{\infty} (a[k]a[\ell]\delta[k-\ell] + d[k]d[\ell]\delta[k-\ell]) \\
&= \sum_{k=-\infty}^{\infty} (|a[k]|^2 + |d[k]|^2).
\end{aligned}$$

Then we have shown the identity with constant $C = 1$.

Problem 2

We treat the symmetric and antisymmetric cases separately. In both cases, the key idea is to factor out the linear-phase term and study the resulting polynomial.

Common factorization Since $h[n]$ has generalized linear phase and length L , we may write

$$H(z) = z^{-(L-1)/2} Q(z),$$

where

$$Q(z) = \sum_{n=0}^{L-1} h[n] z^{-(n-(L-1)/2)}.$$

Apart from trivial zeros at $z = 0$ and $z = \infty$, $H(z)$ and $Q(z)$ share the same zeros. This fact is stated formally in the following proposition.

Proposition 1 (Coincidence of finite zero sets) *Let*

$$H(z) = z^{-\alpha}Q(z), \quad \alpha = \frac{L-1}{2},$$

where $Q(z)$ is a Laurent polynomial. Then the finite zero sets of $H(z)$ and $Q(z)$ coincide, i.e.,

$$\mathcal{Z}_f(H) = \mathcal{Z}_f(Q),$$

where $\mathcal{Z}_f(F)$ denotes the set of finite complex zeros $z \in \mathbb{C} \setminus \{0\}$ such that $F(z) = 0$.

Proof First we show that $\mathcal{Z}_f(H) \supseteq \mathcal{Z}_f(Q)$. Let $z_0 \in \mathcal{Z}_f(Q)$. Then $Q(z_0) = 0$ and $z_0 \neq 0$, and hence

$$H(z_0) = z_0^{-\alpha}Q(z_0) = 0.$$

Thus $z_0 \in \mathcal{Z}_f(H)$.

It remains to show that $\mathcal{Z}_f(H) \subseteq \mathcal{Z}_f(Q)$. Let $z_0 \in \mathcal{Z}_f(H)$. Then $H(z_0) = 0$ with $z_0 \neq 0$, so

$$z_0^{-\alpha}Q(z_0) = 0.$$

Note that for any real α ,

$$z^{-\alpha} \neq 0 \quad \text{for all } z \in \mathbb{C} \setminus \{0\}.$$

Therefore, it follows that $Q(z_0) = 0$, hence $z_0 \in \mathcal{Z}_f(Q)$.

(a) **Symmetric case:** Assume

$$h[n] = h[L-1-n].$$

Then

$$\begin{aligned} Q(z^{-1}) &= \sum_{n=0}^{L-1} h[n]z^{n-(L-1)/2} \\ &= \sum_{n=0}^{L-1} h[L-1-n]z^{n-(L-1)/2} \\ &= Q(z). \end{aligned}$$

Hence, $Q(z)$ is a *palindromic polynomial*, satisfying

$$Q(z^{-1}) = Q(z).$$

If z_0 is a zero of $Q(z)$, then

$$Q(z_0) = 0 \implies Q(z_0^{-1}) = Q(z_0) = 0.$$

Therefore, z_0^{-1} is also a zero of $Q(z)$, and hence of $H(z)$.

(b) **Antisymmetric case:** Assume

$$h[n] = -h[L - 1 - n].$$

Then

$$\begin{aligned} Q(z^{-1}) &= \sum_{n=0}^{L-1} h[n] z^{n-(L-1)/2} \\ &= - \sum_{n=0}^{L-1} h[L - 1 - n] z^{n-(L-1)/2} \\ &= -Q(z). \end{aligned}$$

Thus, $Q(z)$ is an *anti-palindromic polynomial*, satisfying

$$Q(z^{-1}) = -Q(z).$$

If z_0 is a zero of $Q(z)$, then

$$Q(z_0) = 0 \implies Q(z_0^{-1}) = -Q(z_0) = 0.$$

Again, z_0^{-1} is a zero of $Q(z)$, and therefore of $H(z)$.

Problem 3

We argue by contradiction. Assume that $H(z)$ has generalized linear phase.

Step 1: Existence of a nontrivial zero. Since $L \geq 4$, we have

$$\deg R = \frac{L-2}{2} \geq 1.$$

By Lemma 1, $R(z)$ cannot be a power of $(1 + z^{-1})$, and hence $H(z)$ has at least one zero z_0 satisfying

$$z_0 \notin \{\pm 1\}.$$

Step 2: Symmetry of zeros of $P(z) = H(z)H(z^{-1})$. Define

$$P(z) = H(z)H(z^{-1}).$$

Since $P(z)$ is invariant under the transformation $z \mapsto z^{-1}$, we have

$$P(z) = P(z^{-1}).$$

If $H(z_0) = 0$, then

$$P(z_0) = H(z_0)H(z_0^{-1}) = 0,$$

which implies

$$P(z_0^{-1}) = 0.$$

Thus the zeros of $P(z)$ occur in reciprocal pairs $\{z_0, z_0^{-1}\}$.

Step 3: Minimum-phase assignment of zeros. Because $z_0 \notin \{\pm 1\}$, exactly one of $\{z_0, z_0^{-1}\}$ lies strictly inside the unit circle, while the other lies strictly outside. Since $R(z)$ is minimum phase by construction, only the zero inside the unit circle may be assigned to $R(z)$. The reciprocal zero must therefore be assigned to $R(z^{-1})$. Hence, the zero pair $\{z_0, z_0^{-1}\}$ is split asymmetrically between $R(z)$ and $R(z^{-1})$.

Step 4: Contradiction with generalized linear phase. If $H(z)$ had generalized linear phase, then it could be written in the form

$$H(z) = z^{-(L-1)/2}Q(z),$$

where $Q(z)$ has coefficient symmetry or antisymmetry. As shown in Problem 2, this implies that the finite zeros of $H(z)$ occur in reciprocal pairs: if z_0 is a zero of $H(z)$, then z_0^{-1} must also be a zero of $H(z)$ with the same multiplicity. However, the minimum-phase spectral factorization assigns exactly one element of $\{z_0, z_0^{-1}\}$ to $R(z)$ and the other to $R(z^{-1})$. Consequently, $H(z)$ contains only one member of the reciprocal pair as a zero, violating the reciprocal zero symmetry required for generalized linear phase.

Conclusion. This contradiction shows that $H(z)$ cannot satisfy the generalized linear-phase condition. Therefore, an even-length Daubechies orthogonal wavelet filter cannot have generalized linear phase. $\square$